

IoT Vest Tracker System

What it is

The **IoT Vest Tracker System** tracks a vest's location in near real time and raises alerts when the vest's rain/water sensor reports moisture.

The vest sends compact **LoRa** packets to a **Receiver / control tower** (ESP32). The receiver uploads the latest state to **Firestore Realtime Database (RTDB)** when Wi-Fi is available. The **Vest Tracker** Android app (Flutter) reads that data and shows **Live Map**, **Devices**, and **Profile** tabs.

What you use it for (common scenarios)

- **Track a moving vest:** See the latest reported location on the map (and last-seen time on the Devices list).
- **Safety / recovery:** Use the last known point if the vest is misplaced.
- **Weather exposure:** Get a **wet alert** when the sensor transitions to wet so you can dry or protect the vest.

Key features (as delivered)

- **GPS tracking** on an interactive map (OpenStreetMap tiles), with optional focus on one vest ID.
- **LoRa** link from vest to receiver (about **60 s** scheduled updates, plus an immediate transmit on a debounced **dry** → **wet** edge).
- **Rain / water detection** with in-app status and **push-style wet alerts** (foreground listener + optional Android background polling).
- **Multi-device support:** RTDB layout `vest/device_XX` per vest ID; the app lists all devices the signed-in user may read.
- **Malaysia local timestamp** on uploads when using **RECEIVER_CONTROL_TOWER_V2** (`time_my` ISO-8601, UTC+8).

What's included in delivery

- **Vest unit** (transmitter firmware: `firmware/TRANSMITTER_VEST/`)
- **Receiver / control tower** (recommended production sketch: `firmware/RECEIVER_CONTROL_TOWER_V2/`)
- **Mobile app** — Android release APK built from `android-app/vest_tracker/` (Flutter / Firestore)

How it works (short chain)

1. The **vest** reads **GPS** and the **rain sensor**, then sends JSON over **LoRa**.
2. The **receiver** parses packets, optionally **queues** them if the network is down, and **updates** the latest document under `/vest/device_XX` in Firestore RTDB.
3. The **app** (after **Firestore Auth** and correct RTDB rules) streams `vest/` and shows map markers, device cards, and notifications.

Quick start (operators)

1. **Charge** the vest battery pack (see [PowerSafetyNotes.md](#)).
2. **Power on** the vest and the **receiver** (leave the receiver on during use).
3. Install the **APK**, open the app, enter **Firestore RTDB URL** and **Allowed UID** under **Profile**, then **Save settings**.
4. **Sign in** with **email**, **password**, and the **UID** that matches your RTDB rules (see [MobileApp.md](#)).
5. Open **Live Map** and wait for coordinates (GPS works best **outdoors**).

Important limitations

- The **receiver must be on** and within **LoRa range** of the vest for new packets to reach the cloud.
- **GPS** needs a clear view of the sky; indoors or under cover, fixes are slow or absent.
- “**Real-time**” means the **last successful upload**; gaps in Wi-Fi, LoRa, or GPS all delay what you see.

Documentation map

Document	Audience
UserGuide.md	Day-to-day operation
MobileApp.md	Install, permissions, app screens
TroubleshootingGuide.md	Common faults
Hardware.md	Physical parts (non-wiring)
PowerSafetyNotes.md	Battery and safety
SystemArchitecture.md	End-to-end data flow
FirmwareSoftware.md	Payloads, RTDB fields, repo layout
Appendix.md	BOM, pin tables, assembly reference